



Time Allotted : 3 Hours

Full Marks : 70

*The Figures in the margin indicate full marks.**Candidate are required to give their answers in their own words as far as practicable***Group-A (Very Short Answer Type Question)**

1. Answer any ten of the following :

[1 x 10 = 10]

- (i) What type of languages can a Pushdown Automaton accept?
- (ii) What is the "halting state" in a Turing Machine?
- (iii) What does it mean for a problem to be undecidable?
- (iv) What does an edge in a transition diagram represent?
- (v) What type of languages can regular grammar describe?
- (vi) What kind of automaton is used to parse a Context-Free Language?
- (vii) What does a PDA do if the stack is empty during a transition?
- (viii) Can a Turing Machine simulate a finite automaton?
- (ix) Why is state minimization necessary in DFA?
- (x) What is the significance of an initial state in a transition diagram?
- (xi) Write the regular expression for the set of all strings of 0's and 1's beginning with 0 and ending with 1.
- (xii) When is a grammar called type-2 or context-free grammar?

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

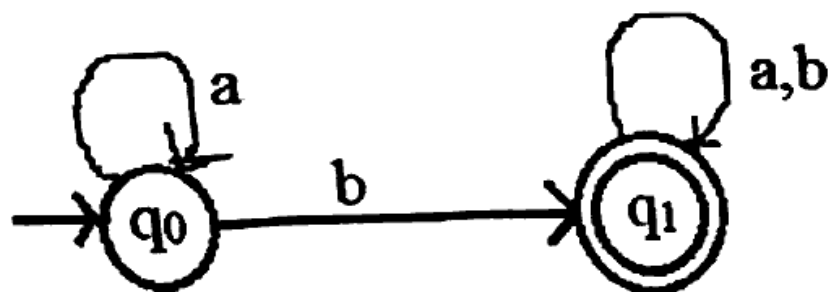
2. Prove $(1+00^*1) + (1+00^*1)(0+10^*1)^*(0+10^*1) = 0^*1(0+10^*1)^*$ [5]
3. If G is the grammar with $P = \{S \rightarrow S+S \mid S^*S \mid a\}$, show that G is ambiguous. [5]
4. Design a pda P accepting the set of all strings over {a, b} with an equal number of a's and b's. [5]
5. Draw the state Transition Diagram of Turing machine for 2's complement of any binary string. [5]
6. Derive the transition diagram for a DFA that accepts strings over {a,b} that have exactly two a's and any number of b's. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. (a) Find a reduced grammar of $P = \{S \rightarrow AB \mid CA, B \rightarrow BC \mid AB, A \rightarrow a, C \rightarrow aB \mid b\}$. [6]
(b) Convert this grammar into Chomsky normal form. $P = \{S \rightarrow aAD, A \rightarrow aB \mid bAB, B \rightarrow b, D \rightarrow d\}$ [5]
(c) Remove NULL productions from this grammar. $P = \{S \rightarrow aS \mid AB, A \rightarrow \wedge, B \rightarrow \wedge, D \rightarrow b\}$ [4]
8. (a) What are the differences between deterministic pushdown automata and non-deterministic pushdown automata? [3]
(b) Design a PDA for the Context Free Language $L = \{a^n b^n \mid n \geq 1\}$. [6]
(c) Construct a pda A equivalent to the following CFG: $P = \{S \rightarrow 0BB, B \rightarrow 0S \mid 1S \mid 0\}$ [6]
9. (a) What is the concept of the tape in a Turing Machine, and how does it function? [4]
(b) Draw the state Transition Diagram of Turing machine for 1's complement of any binary string. [4]
(c) Design a Turing Machine to recognize all strings consisting of even numbers of 1's. [7]
10. (a) What is a Universal Turing Machine (UTM)? [2]
(b) What is the significance of the Halting Problem in computational theory? [3]
(c) Describe how UTM simulates the behavior of any other Turing machine. [5]
(d) What are some common undecidable problems related to formal languages and Turing machines? [5]
11. (a) Write the language L for the grammar $G = (\{S, A\}, \{a, b, c\}, P, S)$ and $P = \{S \rightarrow aS, S \rightarrow A, A \rightarrow bAc \mid bc\}$. [3]
(b) Construct a grammar generating $L = \{a^n b^n c^i \mid n \geq 1, i \geq 0\}$. [4]
(c) Find the Regular Grammar from the DFA given below. [3]



(d) Show that $L = \{a^p \mid p \text{ is a prime}\}$ is not regular.

[5]

*** END OF PAPER ***